Exp15: **Write a program to compute/display dissimilarity matrix (for your own dataset containing at least four instances with two attributes) using Python**

Program:

```
import numpy as np

# Dataset: 4 instances with 2 attributes
data = np.array([
    [1, 2],
    [2, 3],
    [3, 4],
    [5, 6]
])
#------
# Compute dissimilarity (Euclidean distance)
n = len(data)
dissimilarity = np.zeros((n, n))

print(dissimilarity)
#--------


for i in range(n):
    for j in range(n):
        d = np.linalg.norm(data[i] - data[j])  # Euclidean distance
        dissimilarity[i, j] = round(d, 2)


#--------
# Display
print("Dissimilarity Matrix:")
print(dissimilarity)
```

Output:

[[0. 0. 0. 0.]
[0. 0. 0. 0.]
[0. 0. 0. 0.]
[0. 0. 0. 0.]]

Dissimilarity Matrix:
[[0. 1.41 2.83 5.66]
[1.41 0. 1.41 4.24]
[2.83 1.41 0. 2.83]
[5.66 4.24 2.83 0.]]

Exp 16: Write a program to compute/display dissimilarity matrix (for your own dataset containing at least four instances with two attributes) using Python

Program:

program for piechart:

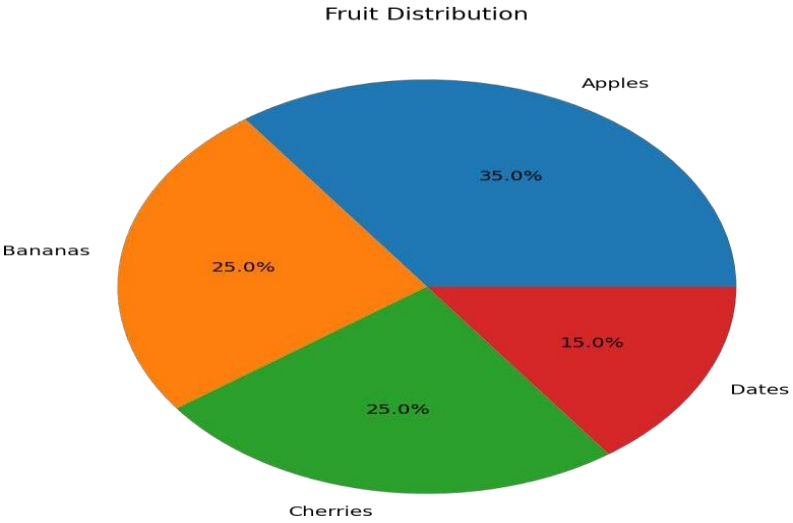
```
import matplotlib.pyplot as plt # Sample data

labels = ['Apples', 'Bananas', 'Cherries', 'Dates'] sizes = [35, 25,25, 15]

# Create the pie chart

plt.pie(sizes, labels=labels, autopct='%1.1f%%') plt.title('FruitDistribution')

plt.show()
```



program for Bargraph:

```
import matplotlib.pyplot as plt import pandas as pd

# Sample data

data = {'Category': ['A', 'B', 'C', 'D'], 'Values': [20, 35, 30, 25]}

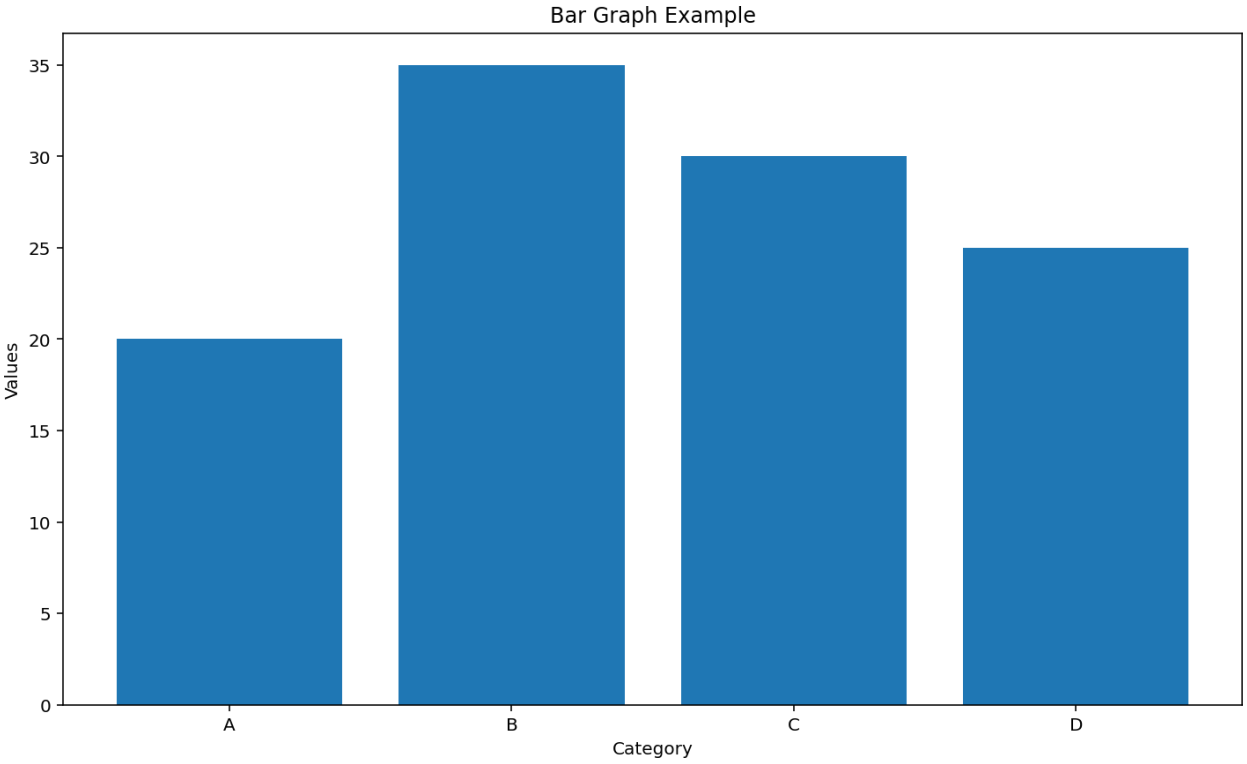
df = pd.DataFrame(data)



# Create a bar graph plt.bar(df['Category'], df['Values']) plt.xlabel('Category') plt.ylabel('Values')

plt.title('Bar Graph Example')
```

plt.show()



program for Barplot:

import matplotlib.pyplot as plt

import numpy as np

Creating dataset

np.random.seed(10)

data = np.random.normal(100, 20, 200)

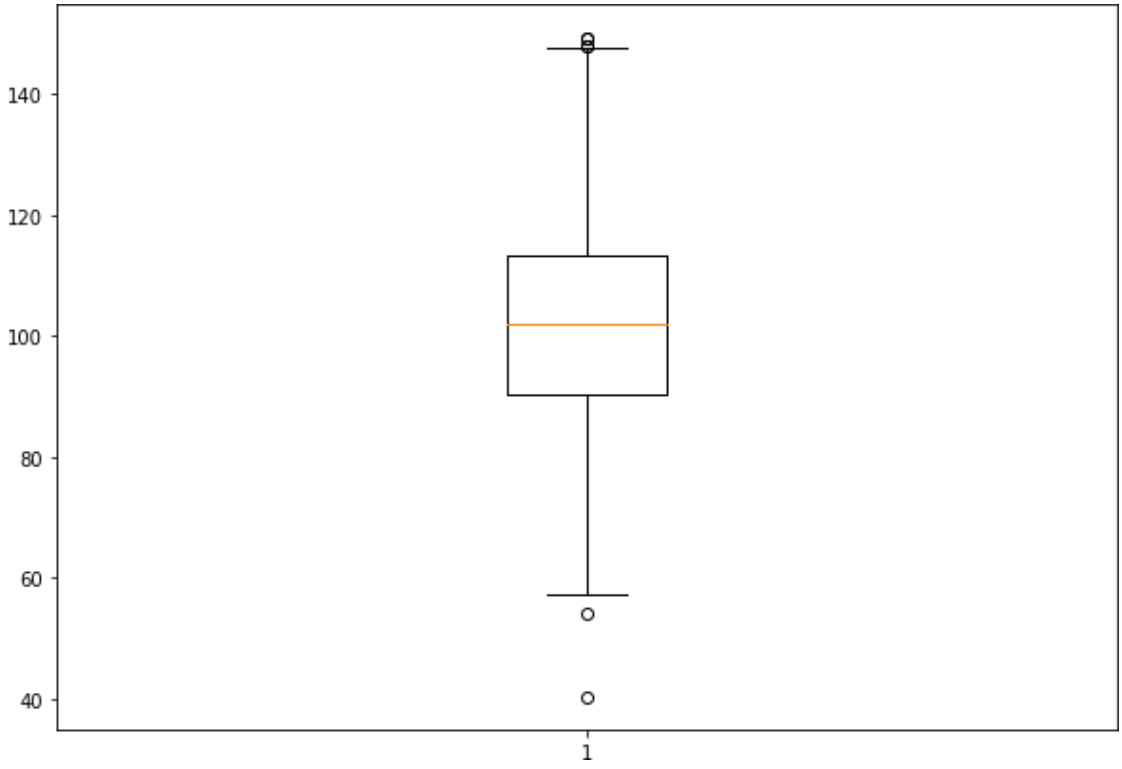
fig = plt.figure(figsize =(10, 7))

Creating plot

plt.boxplot(data)

show plot

plt.show()


140
120
100
80
60
40
1